

The Coset Poincaré Identity for the Top Atom Coefficient of $P_\mu(x; t)$

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Abstract

Let $\mu \vdash n$ and let $P_\mu(x; t)$ denote the Hall–Littlewood polynomial. Expand P_μ in the basis of alternating Demazure atoms A_γ^{alt} (γ ranging over the S_n -orbit of μ): $P_\mu = \sum_\gamma c_\gamma(t) A_\gamma^{\text{alt}}$. We prove

$$c_\mu(t) = \frac{[n]_t!}{\prod_i [m_i(\mu)]_t!} = \binom{n}{m_1(\mu), m_2(\mu), \dots}_t = P(S_n/W_\mu)(t),$$

where $W_\mu = \prod_i S_{m_i(\mu)}$ is the Young-subgroup stabilizer of μ and $\ell(\cdot)$ denotes Bruhat length on $\text{orb}(\mu)$. The proof rests on a *second-moment invariant* of the support of atoms together with an adjoint-pairing identity that isolates c_μ from all other c_γ .

1 Setup

Fix a partition $\mu = (\mu_1 \geq \mu_2 \geq \dots \geq \mu_n \geq 0) \vdash n$ in n parts (with trailing zeros permitted). For each $i \in \{0, 1, 2, \dots\}$ let $m_i(\mu) = |\{j : \mu_j = i\}|$ be the multiplicity of i in μ , and set $v_\mu(t) := \prod_i [m_i(\mu)]_t!$.

Let $\text{orb}(\mu) := \{w \cdot \mu : w \in S_n\}$ be the S_n -orbit of μ (distinct rearrangements of μ), acted on by $(w \cdot \alpha)_j = \alpha_{w^{-1}(j)}$. For $\gamma \in \text{orb}(\mu)$, the Bruhat length $\ell(\gamma)$ is the minimum length of a sequence of adjacent transpositions $s_{i_1}, \dots, s_{i_k}$ (bubble word) such that $s_{i_k} \cdots s_{i_1} \cdot \mu = \gamma$; equivalently, the number of inversions of γ relative to μ .

Define the alternating divided-difference operator

$$\theta_i^{\text{alt}}(f) := \frac{x_{i+1} - tx_i}{x_i - x_{i+1}}(f - s_i f),$$

where s_i swaps $x_i \leftrightarrow x_{i+1}$. For a monomial x^α : if $\alpha_i = \alpha_{i+1}$ then $\theta_i^{\text{alt}}(x^\alpha) = 0$; if $\alpha_i > \alpha_{i+1}$ (write $(a, b) = (\alpha_i, \alpha_{i+1})$) then

$$\theta_i^{\text{alt}}(x^\alpha) = -tx^\alpha + (1-t) \sum_{b < j < a} x_i^j x_{i+1}^{a+b-j} \prod_{k \neq i, i+1} x_k^{\alpha_k} + x^{s_i \alpha}; \quad (1)$$

if $\alpha_i < \alpha_{i+1}$ (write $(b, a) = (\alpha_i, \alpha_{i+1})$) then

$$\theta_i^{\text{alt}}(x^\alpha) = -x^\alpha - (1-t) \sum_{b < j < a} x_i^j x_{i+1}^{a+b-j} \prod_{k \neq i, i+1} x_k^{\alpha_k} + tx^{s_i \alpha}. \quad (2)$$

The atom is defined by

$$A_\gamma^{\text{alt}} := \theta_{i_k}^{\text{alt}} \cdots \theta_{i_1}^{\text{alt}}(x^\mu),$$

where $(i_1, \dots, i_k)$ is a bubble word from μ to γ . In particular $A_\mu^{\text{alt}} = x^\mu$.

Finally, $P_\mu(x; t)$ satisfies $P_\mu = m_\mu + \sum_{\lambda \prec \mu} b_{\lambda, \mu}(t) m_\lambda$ in the monomial basis, where $\prec$ is strict dominance (Macdonald III. §2).

We call an exponent $\alpha \in \mathbb{Z}_{\geq 0}^n$ an *intermediate of x^α at position i* if it agrees with α outside $(i, i+1)$ and its $(i, i+1)$ -entries are a pair $(j, a+b-j)$ with $b < j < a$ where $\{a, b\} = \{\alpha_i, \alpha_{i+1}\}$, $a > b$. Equations (1)–(2) say the support of $\theta_i^{\text{alt}}(x^\alpha)$ is $\{\alpha, s_i \alpha\}$ together with the intermediates of x^α at i .

2 Main theorem

Theorem 1 (Coset Poincaré identity). *For every partition $\mu \vdash n$,*

$$c_\mu(t) = \frac{[n]_t!}{\prod_i [m_i(\mu)]_t!} = \binom{n}{m_1(\mu), m_2(\mu), \dots}_t = P(S_n/W_\mu)(t).$$

The theorem follows from three lemmas, proved in the next three subsections.

2.1 The value $[x^\sigma]P_\mu$ on the orbit

Lemma 2 (Monomial-basis coefficient). *For every $\sigma \in \text{orb}(\mu)$, $[x^\sigma]P_\mu(x; t) = 1$.*

Proof. Write $P_\mu = m_\mu + \sum_{\lambda \prec \mu} b_{\lambda, \mu}(t) m_\lambda$. Since $m_\lambda = \sum_{\tau \in \text{orb}(\lambda)} x^\tau$ and $\text{orb}(\lambda) \cap \text{orb}(\mu) = \emptyset$ for $\lambda \neq \mu$, we have $[x^\sigma]m_\lambda = 0$ for $\lambda \neq \mu$, while $[x^\sigma]m_\mu = 1$ because $\sigma \in \text{orb}(\mu)$. $\square$

2.2 The second-moment invariant

Write $S_2(\alpha) := \sum_j \alpha_j^2$.

Lemma 3 (Second-moment invariant). *For every $\gamma \in \text{orb}(\mu)$ and every $\alpha \in \text{supp}(A_\gamma^{\text{alt}})$,*

$$S_2(\alpha) \leq S_2(\mu).$$

Moreover, S_2 is strictly decreased by every “intermediate” contribution in (1)–(2) and left unchanged by the α -diagonal and $s_i \alpha$ contributions.

Proof. By induction on $\ell(\gamma)$.

Base. $\gamma = \mu$: $\text{supp}(A_\mu^{\text{alt}}) = \{\mu\}$ and $S_2(\mu) \leq S_2(\mu)$. $\checkmark$

Step. $\gamma = s_i \gamma'$ with $\ell(\gamma') = \ell(\gamma) - 1$, so $A_\gamma^{\text{alt}} = \theta_i^{\text{alt}}(A_{\gamma'}^{\text{alt}})$. Fix $\alpha \in \text{supp}(A_\gamma^{\text{alt}})$; then $\alpha \in \text{supp}(\theta_i^{\text{alt}}(x^\beta))$ for some $\beta \in \text{supp}(A_{\gamma'}^{\text{alt}})$ with $\beta_i \neq \beta_{i+1}$.

Case (i): $\alpha = \beta$. Then $S_2(\alpha) = S_2(\beta) \leq S_2(\mu)$ by IH.

Case (ii): $\alpha = s_i \beta$. Same multiset as β , so $S_2(\alpha) = S_2(\beta) \leq S_2(\mu)$.

Case (iii): α is an intermediate of x^β at i . Write $\{a, b\} = \{\beta_i, \beta_{i+1}\}$ with $a > b$, and $(\alpha_i, \alpha_{i+1}) = (j, a+b-j)$ (or the reverse) with $b < j < a$. Then

$$\begin{aligned} S_2(\alpha) - S_2(\beta) &= j^2 + (a+b-j)^2 - a^2 - b^2 \\ &= -[(a^2 + b^2) - (j^2 + (a+b-j)^2)] \\ &= -2(a-j)(j-b) < 0 \end{aligned}$$

because $b < j < a$. So $S_2(\alpha) < S_2(\beta) \leq S_2(\mu)$. $\square$

The last identity— $(a^2 + b^2) - (j^2 + (a+b-j)^2) = 2(a-j)(j-b)$ —is the crucial arithmetic fact. It says: replacing an outer pair $\{a, b\}$ by an inner pair $\{j, a+b-j\}$ with the same sum *strictly reduces* the sum of squares.

2.3 The Weyl-symmetrization identity

Define the linear functional $L : \mathbb{Z}[t][x_1, \dots, x_n] \rightarrow \mathbb{Z}[t]$ by

$$L(f) := \sum_{\sigma \in \text{orb}(\mu)} t^{\ell(\sigma)} [x^\sigma] f.$$

Equivalently, $L(f) = \langle f, Q \rangle$ where $Q := \sum_{\sigma \in \text{orb}(\mu)} t^{\ell(\sigma)} x^\sigma$ and $\langle x^\alpha, x^\beta \rangle := \delta_{\alpha, \beta}$ is the standard monomial pairing.

Lemma 4 (Weyl-symmetrization identity). *For every $\gamma \in \text{orb}(\mu)$,*

$$L(A_\gamma^{\text{alt}}) = \delta_{\gamma, \mu}.$$

Proof. Induction on $\ell(\gamma)$.

Base. $\gamma = \mu$: $A_\mu^{\text{alt}} = x^\mu$, so $L(A_\mu^{\text{alt}}) = t^{\ell(\mu)} = t^0 = 1$. $\checkmark$

Step. $\gamma = s_i \gamma'$ with $\ell(\gamma') < \ell(\gamma)$. We show $L(\theta_i^{\text{alt}}(A_{\gamma'}^{\text{alt}})) = 0$.

Let $\tilde{\theta}_i$ denote the adjoint of θ_i^{alt} with respect to $\langle \cdot, \cdot \rangle$: $\langle \theta_i^{\text{alt}} f, g \rangle = \langle f, \tilde{\theta}_i g \rangle$, i.e. $[x^\alpha] \tilde{\theta}_i(x^\beta) = [x^\beta] \theta_i^{\text{alt}}(x^\alpha)$. Then

$$L(\theta_i^{\text{alt}}(A_{\gamma'}^{\text{alt}})) = \langle \theta_i^{\text{alt}}(A_{\gamma'}^{\text{alt}}), Q \rangle = \langle A_{\gamma'}^{\text{alt}}, \tilde{\theta}_i(Q) \rangle.$$

We evaluate $\tilde{\theta}_i(Q)$ and pair it against $A_{\gamma'}^{\text{alt}}$.

Step (a): the orbit-supported part of $\tilde{\theta}_i(Q)$ vanishes.

Reading off (1)–(2), for each $\sigma \in \text{orb}(\mu)$ the “diagonal-and-swap” contribution to $\tilde{\theta}_i(x^\sigma)$ is

$$\begin{cases} -t x^\sigma + t x^{s_i \sigma} & \text{if } \sigma_i > \sigma_{i+1} \text{ (descent),} \\ -x^\sigma + x^{s_i \sigma} & \text{if } \sigma_i < \sigma_{i+1} \text{ (ascent),} \\ 0 & \text{if } \sigma_i = \sigma_{i+1}. \end{cases}$$

Indeed, $[x^\sigma] \theta_i^{\text{alt}}(x^\sigma)$ contributes at $\alpha = \sigma$ (giving $-t$ or -1 or 0), and $[x^\sigma] \theta_i^{\text{alt}}(x^{s_i \sigma})$ contributes at $\alpha = s_i \sigma$ (giving t or 1).

Descent-ascent pairs $(\sigma, s_i \sigma)$ in the orbit satisfy $\ell(s_i \sigma) = \ell(\sigma) + 1$ when σ has a descent at $(i, i+1)$ (swapping a descent up the Bruhat order). Grouping σ (descent) with $\sigma' := s_i \sigma$ (ascent), the diagonal-and-swap contributions to $\tilde{\theta}_i(Q) = \sum_{\sigma} t^{\ell(\sigma)} \tilde{\theta}_i(x^\sigma)$ at x^σ and $x^{\sigma'}$ are:

$$\begin{aligned} [x^\sigma] : \quad & t^{\ell(\sigma)}(-t) + t^{\ell(\sigma')}(1) = -t^{\ell(\sigma)+1} + t^{\ell(\sigma)+1} = 0, \\ [x^{\sigma'}] : \quad & t^{\ell(\sigma)}(t) + t^{\ell(\sigma')}(-1) = t^{\ell(\sigma)+1} - t^{\ell(\sigma)+1} = 0. \end{aligned}$$

Orbit elements fixed by s_i contribute 0 directly. Hence the orbit-supported part of $\tilde{\theta}_i(Q)$ vanishes.

Step (b): the non-orbit part of $\tilde{\theta}_i(Q)$ has strictly higher second moment.

The remaining contribution to $\theta_i(x^\sigma)$, for each $\sigma \in \text{orb}(\mu)$, comes from “reverse intermediates”: $\alpha \neq \sigma, s_i \sigma$ such that σ is an intermediate of $\theta_i^{\text{alt}}(x^\alpha)$. Reading off (1)–(2), such α has $(\alpha_i, \alpha_{i+1}) = (u, v)$ or (v, u) with $u > v$, $u + v = \sigma_i + \sigma_{i+1}$, $v < \min(\sigma_i, \sigma_{i+1})$, $u > \max(\sigma_i, \sigma_{i+1})$, and $\alpha_k = \sigma_k$ for $k \neq i, i+1$. Set $M := \max(\sigma_i, \sigma_{i+1})$ and $M' := \min(\sigma_i, \sigma_{i+1})$ (so $M + M' = u + v$), and write $p := u - M \geq 1$ (integer). Then $v = M' - p$, and the constraint $v \geq 0$ says $p \leq M'$. Consequently,

$$\begin{aligned} S_2(\alpha) - S_2(\sigma) &= (u^2 + v^2) - (M^2 + M'^2) \\ &= (M + p)^2 + (M' - p)^2 - M^2 - M'^2 \\ &= 2p(M - M') + 2p^2 \geq 2p^2 \geq 2 > 0 \end{aligned}$$

(using $M \geq M'$ and $p \geq 1$). Since $\sigma \in \text{orb}(\mu)$ implies $S_2(\sigma) = S_2(\mu)$, every reverse-intermediate α satisfies

$$S_2(\alpha) > S_2(\mu). \quad (3)$$

Step (c): the pairing vanishes.

By Lemma 3, every $\alpha \in \text{supp}(A_{\gamma'}^{\text{alt}})$ satisfies $S_2(\alpha) \leq S_2(\mu)$. In particular $[x^\alpha]A_{\gamma'}^{\text{alt}} = 0$ for every α with $S_2(\alpha) > S_2(\mu)$. By step (a), the orbit-supported part of $\tilde{\theta}_i(Q)$ vanishes; and by step (b) plus (3), every non-orbit monomial in the support of $\tilde{\theta}_i(Q)$ has $S_2(\alpha) > S_2(\mu)$, hence pairs to zero against $A_{\gamma'}^{\text{alt}}$. Therefore

$$L(A_{\gamma'}^{\text{alt}}) = \langle A_{\gamma'}^{\text{alt}}, \tilde{\theta}_i(Q) \rangle = 0,$$

completing the induction. $\square$

2.4 Proof of Theorem 1

Applying L to $P_\mu = \sum_{\gamma \in \text{orb}(\mu)} c_\gamma(t) A_\gamma^{\text{alt}}$ and using Lemma 4:

$$L(P_\mu) = \sum_{\gamma} c_\gamma(t) L(A_\gamma^{\text{alt}}) = c_\mu(t).$$

On the other hand, using Lemma 2,

$$L(P_\mu) = \sum_{\sigma \in \text{orb}(\mu)} t^{\ell(\sigma)} [x^\sigma] P_\mu = \sum_{\sigma \in \text{orb}(\mu)} t^{\ell(\sigma)} = P(S_n/W_\mu)(t).$$

The last equality is standard: the parabolic Poincaré polynomial $P(S_n/W_\mu)(t) = \sum_{w \in D_\mu} t^{\ell(w)}$, where D_μ is the set of minimum-length coset representatives for S_n/W_μ ; the map $w \mapsto w \cdot \mu$ is a bijection $D_\mu \rightarrow \text{orb}(\mu)$ preserving length. Finally,

$$P(S_n/W_\mu)(t) = \binom{n}{m_1(\mu), m_2(\mu), \dots}_t = \frac{[n]_t!}{\prod_i [m_i(\mu)]_t!}.$$

Combining, $c_\mu(t) = [n]_t! / v_\mu(t)$. $\square$

3 Verification

3.1 Computational evidence

Verified numerically (via SageMath and SymPy scripts) on:

- Every partition of n with $n \leq 5$ that has been tested previously (12 partitions: $(2, 1, 0)$, $(2, 2, 0)$, $(2, 2, 1)$, $(3, 2, 1)$, $(3, 3, 1)$, $(4, 1, 1, 1)$, $(4, 2, 1, 0)$, $(4, 3, 2, 1, 0)$, $(4, 2, 2, 0)$ checking that (i) the atom-triangularity lemma holds, (ii) the second-moment invariant holds (equality attained only at $\alpha = \sigma \in \text{orb}(\mu)$), (iii) the Weyl-symmetrization identity $L(A_\gamma^{\text{alt}}) = \delta_{\gamma, \mu}$ holds, (iv) the resulting c_μ matches $[n]_t! / v_\mu(t)$.

Zero failures across all cases.

3.2 Worked example: $\mu = (2, 2, 0)$

Orbit is $\{(2, 2, 0), (2, 0, 2), (0, 2, 2)\}$ with Bruhat lengths 0, 1, 2.

Atoms. By direct computation via (1):

$$A_{(2,2,0)}^{\text{alt}} = x_1^2 x_2^2,$$

$$A_{(2,0,2)}^{\text{alt}} = -t x_1^2 x_2^2 + (1-t) x_1^2 x_2 x_3 + x_1^2 x_3^2,$$

$$A_{(0,2,2)}^{\text{alt}} = -t(1-t) x_1^2 x_2 x_3 + (1-t) x_1 x_2^2 x_3 - t x_1^2 x_3^2 + (1-t) x_1 x_2 x_3^2 + x_2^2 x_3^2.$$

Second moments:

- $A_{(2,2,0)}^{\text{alt}}$: only $S_2 = 8 = S_2(\mu)$.
- $A_{(2,0,2)}^{\text{alt}}$: monomials $(2, 2, 0), (2, 1, 1), (2, 0, 2)$ have $S_2 = 8, 6, 8$; all ≤ 8 .
- $A_{(0,2,2)}^{\text{alt}}$: monomials $(2, 1, 1), (1, 2, 1), (2, 0, 2), (1, 1, 2), (0, 2, 2)$ have $S_2 = 6, 6, 8, 6, 8$; all ≤ 8 .

The identity $L(A_\gamma^{\text{alt}}) = \delta_{\gamma,\mu}$:

$$L(A_{(2,2,0)}^{\text{alt}}) = t^0 \cdot 1 = 1.$$

$$L(A_{(2,0,2)}^{\text{alt}}) = t^0(-t) + t^1(0) + t^2(0) + (\text{non-orbit terms} \rightarrow 0)$$

[explicitly: only $(2, 2, 0)$ and $(2, 0, 2)$ are in orbit:]

$$= 1 \cdot (-t) + t \cdot 1 = 0.$$

$$L(A_{(0,2,2)}^{\text{alt}}) = t^0 \cdot 0 + t^1(-t) + t^2 \cdot 1 = -t^2 + t^2 = 0.$$

Verification of c_μ : solving $c_\gamma \cdot M = (1, 1, 1)$ by back-substitution (with M upper triangular, $M[\gamma, \sigma] = [x^\sigma]A_\gamma^{\text{alt}}$, ordered by decreasing ℓ): $c_{(0,2,2)} = 1$, $c_{(2,0,2)} = 1 + t = [2]_t$, $c_{(2,2,0)} = 1 + t + t^2 = [3]_t$. And indeed $c_\mu = [3]_t = [3]_t! / (v_\mu = [2]_t) = [3]_t! / [2]_t \cdot [1]_t = [3]_t$. ✓

4 Remarks

Remark 5. The proof uses only the following inputs:

1. The definition of θ_i^{alt} and the associated atom construction.
2. The monomial expansion $P_\mu = m_\mu +$ strictly-lower-dominance terms.
3. The parabolic Poincaré polynomial identity $\sum_{w \in D_\mu} t^{\ell(w)} = [n]_t! / v_\mu(t)$ (Chevalley–Shephard–Todd).

The two novel structural facts—the second-moment invariant (Lemma 3) and the Weyl-symmetrization identity (Lemma 4)—may be of independent interest.

Remark 6. Lemma 4 is strictly stronger than the top-coefficient identity: it gives a *characterization* of the top atom by a linear functional. In particular, L picks out c_μ exactly, while all other c_γ are annihilated. This suggests L may extend to a duality between the atom basis and a suitably-graded orbit basis, which we intend to explore.

Remark 7. The second-moment invariant reflects a general geometric fact: the operation of “pulling in” an outer pair (a, b) with $a > b$ to an inner pair $(j, a + b - j)$ with $b < j < a$ is strictly a convex-combination move (it decreases the “spread” of the multiset while preserving the mean). The formal identity $(a^2 + b^2) - (j^2 + (a + b - j)^2) = 2(a - j)(j - b)$ is thus a manifestation of convexity. This ties the atom-support structure to a classical inequality.